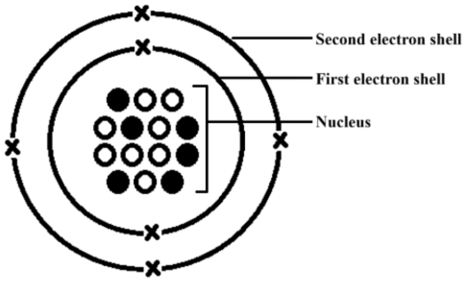
Intensive notes (Topic 2: Microscopic world I)

Atom

Atom is the smallest part of an element which has the chemical properties of that element.

Subatomic particle	Proton	Neutron	Electron
Relative mass	1	1	Negligible
Charge	+1	0	-1
Position	Nucleus	Nucleus	Moving around nucleus



Note 1: An atom is mostly empty space. **Mass is concentrated in the nucleus** of an atom.

Note 2: Protons and electrons exist in ALL atoms, but not neutrons (e.g. $^{1}_{1}H$ atom does not have neutrons)

Note 3: Atoms are **electrically neutral** because the **number of positively charged protons** is equal to the **number of negatively charged electrons**

Atomic symbol

Atomic number (No. of protons)	Atom	Atomic symbol	Electronic arrangement	Atomic number (No. of protons)	Atom	Atomic symbol	Electronic arrangement
1	Hydrogen*	H	1	11	Sodium#	Na	2,8,1
2	Helium*	He	2	12	Magnesium#	Mg	2,8,2
3	Lithium#	Li	2,1	13	Aluminium#	Al	2,8,3
4	Beryllium#	Be	2,2	14	Silicon^	Si	2,8,4
5	Boron^	B	2,3	15	Phosphorus*	P	2,8,5
6	Carbon*	C	2,4	16	Sulphur*	S	2,8,6
7	Nitrogen*	N	2,5	17	Chlorine*	Cl	2,8,7
8	Oxygen*	O	2,6	18	Argon*	Ar	2,8,8
9	Fluorine*	F	2,7	19	Potassium#	K	***2,8,8,1***
10	Neon*	Ne	2,8	20	Calcium#	Ca	***2,8,8,2***

#: Metal ^: Semi-metal *: Non-metal

Note 1: Top 5 most abundant elements in Earth crust are O > Si > Al > Fe > Ca.

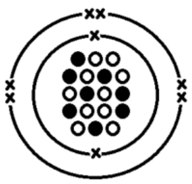
Note 2: Maximum number of electrons a shell can hold can be calculated by $2n^2$, where n is the shell number. Therefore the outermost shell of Ar (2,8,8) is NOT FULLY filled (3rd shell can hold 18 electrons)

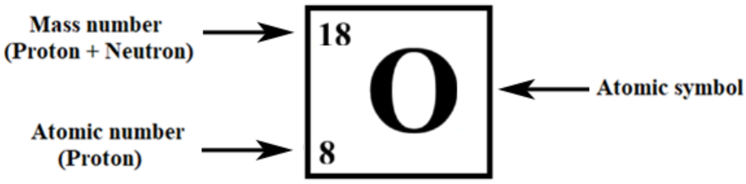
Full atomic symbol

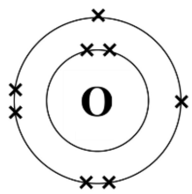
Cross = electron (Cross moves around nucleus)

Black dot = proton (Black dot is inside the nucleus, and the numbers of black dot and cross are equal)

White dot = neutron (White dot is inside the nucleus)







Electron diagram of an oxygen atom

Intensive notes (Topic 2: Microscopic world I)

Isotopes

Isotopes are **atoms** of the element with **same number of protons** but **different number of neutrons**	n = 7 p = 6	n = 8 p = 6
Relative atomic mass is the **weighted average** of the relative isotopic masses of all of its naturally occurring isotopes of the ^{12}C = 12.00 scale	**^{13}C atom** (6 protons, 7 neutrons, 6 electrons)	**^{14}C atom** (6 protons, 8 neutrons, 6 electrons)

- Mass of a ^{12}C atom = 1.9926 x 10^{-23} g Relative isotopic mass of a ^{12}C atom = 12.00
- Mass of a ^{24}Mg atom = 3.9828 x 10^{-23} g Relative isotopic mass of a ^{24}Mg atom = 23.98 (= 12 x 3.9828/1.9926)
- Mass of a ^{27}Al atom = 4.4783 x 10^{-23} g Relative isotopic mass of a ^{27}Al atom = 26.98 (= 12 x 4.4783/1.9926)

On the ^{12}C = 12.00 scale, the relative isotopic mass of an isotope is roughly equal to its mass number. It carries **no unit**.

Note : Isotopes have **SAME chemical properties** (NOT SIMILAR) because they are **atoms with same electronic arrangement**. They CANNOT be separated by chemical methods. In addition, isotopes have **slightly different physical properties** (e.g. mass). Therefore they CAN be separated by physical methods.

Calculation related to relative atomic mass

Example 1: Chlorine occurs naturally in two stable isotopes, ^{35}Cl and ^{37}Cl. If the abundance of ^{35}Cl is 75.8%, calculate the relative atomic mass of Cl.

Solution: Note that the abundance of ^{37}Cl is 100% – 75.8% = 24.2%

$$\mathrm{R.A.M.} = \frac{(35)(75.8) + (37)(24.2)}{100}$$

$$\mathrm{R.A.M.} = 35.484$$

Example 2: Magnesium occurs naturally in three stable isotopes, ^{24}Mg, ^{25}Mg and ^{26}Mg. If the abundance of ^{24}Mg is 78.6% and the relative atomic mass of Mg is 24.327, calculate the abundance of ^{25}Mg.

Solution: Let x% be the abundance of ^{25}Mg while y% be the abundance of ^{26}Mg

Since 78.6 + x + y = 100

y = 21.4 – x

$$24.327 = \frac{(24)(78.6) + (25)(x) + (26)(21.4 - x)}{100}$$

$$x = 10.1 \text{ (The abundance of } ^{25}\text{Mg is 10.1\%)}$$

Example 3: Bromine occurs naturally in two stable isotopes, ^{79}Br and ^{x}Br. If the abundance of ^{79}Br is 50.6% and the relative atomic mass of Br is 79.988, calculate the isotopic mass of ^{x}Br.

Solution: Note that the abundance of ^{x}Br is 100% – 50.6% = 49.4%

$$79.988 = \frac{(79)(50.6) + (x)(49.4)}{100}$$

$$x = 81 \text{ (The isotopic mass of } ^{x}\text{Br is 81)}$$

Intensive notes (Topic 2: Microscopic world I)

Classification of elements

Property	Metal	Non-metal
State	Solid (Only liquid: Mercury, $_{80}$Hg)	Solid or gas (Only liquid: Bromine $_{35}$Br)
Appearance	Shiny	Dull
m.p. / b.p.	Usually high (except Alkali Metals and mercury)	Usually low (Except graphite and diamond)
Hardness	Usually hard and strong	Usually brittle
Malleable / Ductile	Yes	No
Density	Usually high	Usually low
Electrical Conductivity	Good conductor	Poor conductor (Except graphite)

Note: Boron and silicon are classified as **metalloid / semi-metal.** They have properties of both metals and non-metals. They conduct electricity under high temperature / with impurities. (Silicon is used in making computer chips)

Periodic Table

- In modern Periodic Table, elements are arranged in **increasing atomic number. (NOT mass number)**

  Horizontal row: **Period** = Number of occupied electron shells

  Vertical column: **Group** = Number of electrons in outermost shell
- Across a period, elements change from metals through semi-metal to non-metals.

I	II											III	IV	V	VI	VII	0
				Atomic number →	$_{1}$H 1	← Electronic arrangement											$_{2}$He 2
$_{3}$Li 2,1	$_{4}$Be 2,2											$_{5}$B 2,3	$_{6}$C 2,4	$_{7}$N 2,5	$_{8}$O 2,6	$_{9}$F 2,7	$_{10}$Ne 2,8
$_{11}$Na 2,8,1	$_{12}$Mg 2,8,2	**Transition metals**										$_{13}$Al 2,8,3	$_{14}$Si 2,8,4	$_{15}$P 2,8,5	$_{16}$S 2,8,6	$_{17}$Cl 2,8,7	$_{18}$Ar 2,8,8
$_{19}$K 2,8,8,1	$_{20}$Ca 2,8,8,2	$_{21}$Sc 2,8,9,2	$_{22}$Ti 2,8,10,2	$_{23}$V 2,8,11,2	$_{24}$Cr 2,8,13,1	$_{25}$Mn 2,8,13,2	$_{26}$Fe 2,8,14,2	$_{27}$Co 2,8,15,2	$_{28}$Ni 2,8,16,2	$_{29}$Cu 2,8,18,1	$_{30}$Zn 2,8,18,2	$_{31}$Ga 2,8,18,3	$_{32}$Ge 2,8,18,4	$_{33}$As 2,8,18,5	$_{34}$Se 2,8,18,6	$_{35}$Br 2,8,18,7	$_{36}$Kr 2,8,18,8
$_{37}$Rb 2,8,18,8,1	$_{38}$Sr 2,8,18,8,2															$_{53}$I 2,8,18,18,7	$_{54}$Xe 2,8,18,18,8
$_{55}$Cs 2,8,18,18,8,1	$_{56}$Ba 2,8,18,18,8,2															$_{85}$At 2,8,18,32,18,7	$_{86}$Rn 2,8,18,32,18,8

Note 1: Hydrogen does NOT belong to any group

Note 2: In Group 0, **Helium has 2** outermost shell electrons while **other elements have 8**

Note 3: Elements between Group II and Group III are **transition metals**

Note 4: Elements in the same group have **SIMILAR chemical properties** (NOT SAME) because they have **same number of outermost shell electrons.**

Group I: Alkali metals

- Soft metals (can be cut by a knife) with low densities (float on water)
- Very reactive (store in paraffin oil to prevent reaction with oxygen)
- React with water to give metal hydroxide (alkaline solution) and hydrogen gas

  Word equation for the reaction between sodium and water:

  Sodium + Water ➔ Sodium hydroxide + Hydrogen
- Reactivity of Group I elements **increases down the group.**

Element	Reaction with water
Lithium (Li)	Floats, giving off hydrogen **steadily**
Sodium (Na)	Melts to form a silvery ball which **moves quickly** on the water surface Giving off hydrogen **rapidly** The metal burns with a **golden yellow flame**
Potassium (K)	Melts to form a silvery ball which **moves very quickly** on the water surface Giving off hydrogen **very rapidly**. The metal burns with a **lilac flame**.
Rubidium (Rb)	Reacts even **more vigorously than potassium** does. Explosion may occur
Caesium (Cs)	Reacts even **more vigorously than rubidium** does. Explosion may occur

Group II: Alkaline earth metals

- All have low densities. However, they are denser than Group I elements of the same period.
- All are reactive metals. However, they are less reactive than Group I elements of the same period
- React with water less vigorously than Group I elements to give metal hydroxide and hydrogen (except Be)

  Word equation for the reaction between calcium and water:

  Calcium + Water ➔ Calcium hydroxide + Hydrogen
- Reactivity of Group II elements **increases down the group.**

Element	Reaction with water
Beryllium (Be)	**Does not react** with water or steam
Magnesium (Mg)	Has **almost no reaction** with cold water but reacts with steam to produce hydrogen
Calcium (Ca)	Reacts **steadily** with cold water to produce hydrogen
Strontium (Sr)	Reacts **vigorously** with cold water
Barium (Ba)	Reacts **even more vigorously than strontium** does

- React with hydrochloric acid to give metal chloride and hydrogen

  Word equation for the reaction between magnesium and hydrochloric acid:

  Magnesium + Hydrochloric acid ➔ Magnesium chloride + Hydrogen
- React with non-metals to form ionic compounds (except Be)

  Word equation for the reaction between magnesium and chlorine:

  Magnesium + Chlorine ➔ Magnesium chloride

Group VII: Halogens

- All are colored
- Melting points and boiling points of the Group VII elements **increase down the group**

	Fluorine (F)	Chlorine (Cl)	Bromine (Br)	Iodine (I)	Astatine (At)
Color in solid state	---	---	---	Black*	Black*
Color in liquid state	---	---	Reddish brown	---	---
Color in gas state	Yellow*	Yellowish green*	Reddish brown	Purple	---
Color in aqueous state	---	Pale green	Brown	Brown	---

*Normal state under room conditions

- All are toxic
- All react with metals to give ionic compounds

  Word equation for the reaction between sodium and bromine:

  Sodium + Bromine ➔ Sodium bromide
- All react with non-metals to give covalent compounds

  Word equation for the reaction between hydrogen and chlorine:

  Hydrogen + Chlorine ➔ Hydrogen chloride
- Reactivity of Group VII elements **decreases down the group.**

Element	Reaction with hydrogen
Fluorine (F)	Reacts **explosively** even in the **dark**
Chlorine (Cl)	Reacts **explosively in sunlight**, but the reaction is **slow in the dark**
Bromine (Br)	Reacts **only in sunlight** or when **heated**
Iodine (I)	Has **almost no reaction** even in direct sunlight or upon strong heating

Group 0: Noble gases

- All are colorless gas at room conditions
- All are **very unreactive**, They have little or no reaction with other elements. (May have reactions under extreme conditions)

  Noble gases are **unreactive / inert** since they have an **octet structure (duplet in helium)** in their outermost shell.

Element	Electronic arrangement	Use
Helium (He)	2	Fill balloons and airships (Helium has low density and does not burn)
Neon (Ne)	2, 8	Fill advertising signs (Neon does not react with glass tube)
Argon (Ar)	2, 8, 8	Fill light bulbs (Argon does not react with the metal filament in the bulb)

Intensive notes (Topic 2: Microscopic world I)

Ionic bond

Ionic bond is a strong, non-directional electrostatic attraction between oppositely charged ions. It is formed by the transfer of one or more electrons from one atom to another

Formation of ion: Atoms tend to attain the electronic arrangement of the nearest noble gas (stable octet/duplet structure) by gaining or losing electrons (metal) to form an ion.

Group	I	II	III	IV	V	VI	VII	0
Charge of stable ion	+	2+	3+	------	3–	2–	–	------

- Atoms of metals tend to lose electrons while atoms of non-metals tend to gain electrons.

$$Na \longrightarrow Na^{+} + e^{-}$$

Sodium atom (11 protons + 11 electrons) → Sodium ion (11 protons + 10 electrons)

$$Al \longrightarrow Al^{3+} + 3e^{-}$$

Aluminium atom (13 protons + 13 electrons) → Aluminium ion (13 protons + 10 electrons)

$$O + 2e^{-} \longrightarrow O^{2-}$$

Oxygen atom (8 protons + 8 electrons) → Oxide ion (8 protons + 10 electrons)

$$Cl + e^{-} \longrightarrow Cl^{-}$$

Chlorine atom (17 protons + 17 electrons) → Chloride ion (17 protons + 18 electrons)

Steps for drawing electron diagrams of ionic compounds

Step 1: Determine the charge of the ions

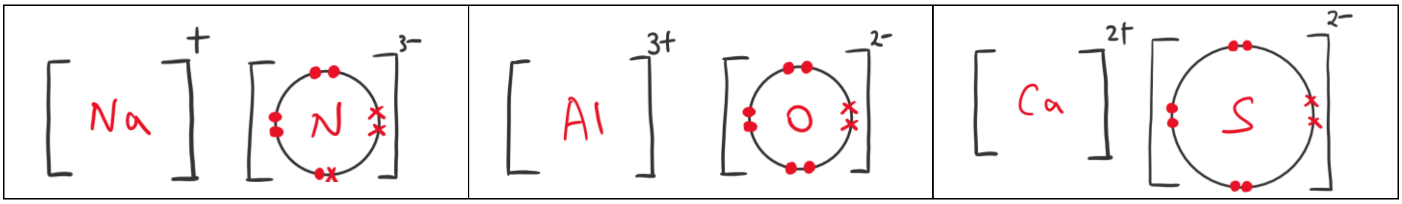
Example 1: Na + N	Example 2: Al + O	Example 3: Ca + S
$Na \rightarrow Na^{+}$ $N \rightarrow N^{3-}$	$Al \rightarrow Al^{3+}$ $O \rightarrow O^{2-}$	$Ca \rightarrow Ca^{2+}$ $S \rightarrow S^{2-}$

Step 2: Write down the chemical formula of the ionic compound formed

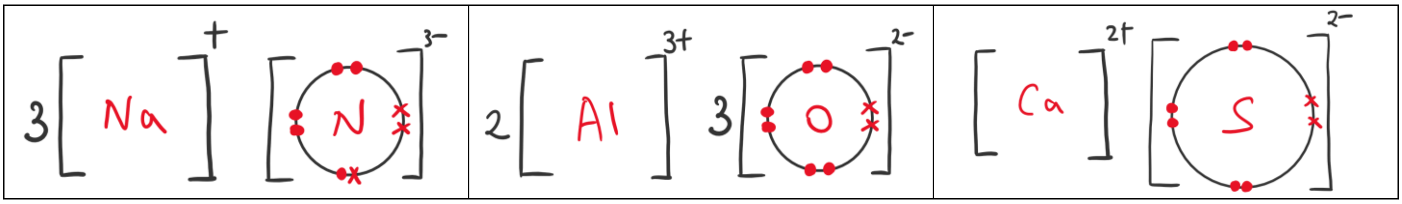
Na^{1+} N^{3-} → Na_3 N_1	Al^{3+} O^{2-} → Al_2 O_3	Ca^{2+} S^{2-} → Ca_2 S_2
Chemical formula: Na_3N (Not Na_3N_1)	Chemical formula: Al_2O_3	Chemical formula: CaS (Not Ca_2S_2)

Step 3: Draw the electron diagram for cation and anion

In most cases, cations do not have outermost shell electrons while anions have 8 outermost shell electrons (except H^{-})



Step 4: Fill in the number of ions base on the formula determined in step 2



Intensive notes (Topic 2: Microscopic world I)

Formulae and names of ions

Cation (Positively charged ion)

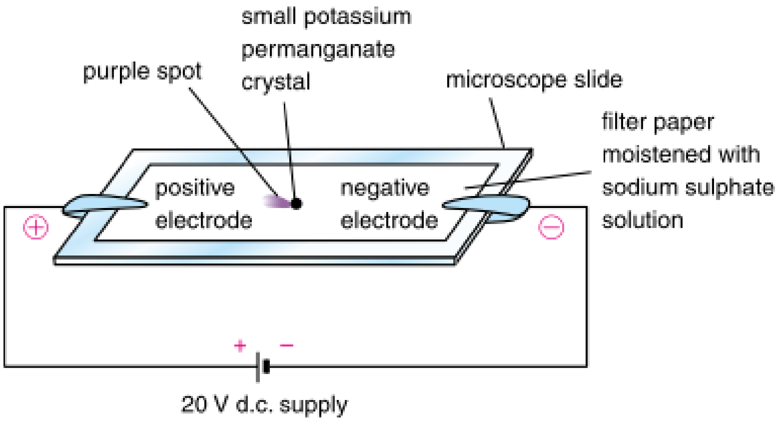
Charge	Formula	Name	(Color)
1+	Na^{+}	Sodium ion	
	K^{+}	Potassium ion	
	H^{+}	**Hydrogen ion**	
	NH_4^{+}	**Ammonium ion**	
	Ag^{+}	**Silver ion**	
	Cu^{+}	Copper(I) ion	
	Hg^{+}	Mercury(I) ion	
2+	Mg^{2+}	Magnesium ion	
	Ca^{2+}	Calcium ion	
	Ba^{2+}	Barium ion	
	Zn^{2+}	**Zinc ion**	
	Pb^{2+}	**Lead(II) ion**	
	Mn^{2+}	Manganese(II) ion	***(Pale pink)***
	Fe^{2+}	Iron(II) ion	***(Green)***
	Co^{2+}	Cobalt(II) ion	***(Pink)***
	Ni^{2+}	Nickel(II) ion	***(Green)***
	Cu^{2+}	Copper(II) ion	***(Blue)***
	Hg^{2+}	Mercury(II) ion	
3+	Al^{3+}	Aluminium ion	
	Fe^{3+}	Iron(III) ion	***(Yellow)***
	Cr^{3+}	Chromium(III) ion	***(Green)***

Anion (Negatively charged ion)

Charge	Formula	Name	(Color)
1-	H^{-}	**Hydride ion**	
	OH^{-}	Hydroxide ion	
	Cl^{-}	Chloride ion	
	Br^{-}	Bromide ion	
	I^{-}	Iodide ion	
	NO_3^{-}	Nitrate ion	
	NO_2^{-}	Nitrite ion	
	HCO_3^{-}	Hydrogencarbonate ion	
	HSO_4^{-}	Hydrogensulphate ion	
	CN^{-}	Cyanide ion	
	ClO_3^{-}	Chlorate ion	
	ClO^{-}	Hypochlorite ion	
	MnO_4^{-}	Permanganate ion	***(Purple)***
2-	O^{2-}	Oxide ion	
	S^{2-}	Sulphide ion	
	SO_4^{2-}	Sulphate ion	
	SO_3^{2-}	Sulphite ion	
	$S_2O_3^{2-}$	Thiosulphate ion	
	CO_3^{2-}	Carbonate ion	
	SiO_3^{2-}	Silicate ion	
	CrO_4^{2-}	Chromate ion	***(Yellow)***
	$Cr_2O_7^{2-}$	Dichromate ion	***(Orange)***
3-	N^{3-}	Nitride ion	
	P^{3-}	Phosphide ion	
	PO_4^{3-}	Phosphate ion	

Experiments showing migration of ion

Experiment 1: Using filter paper



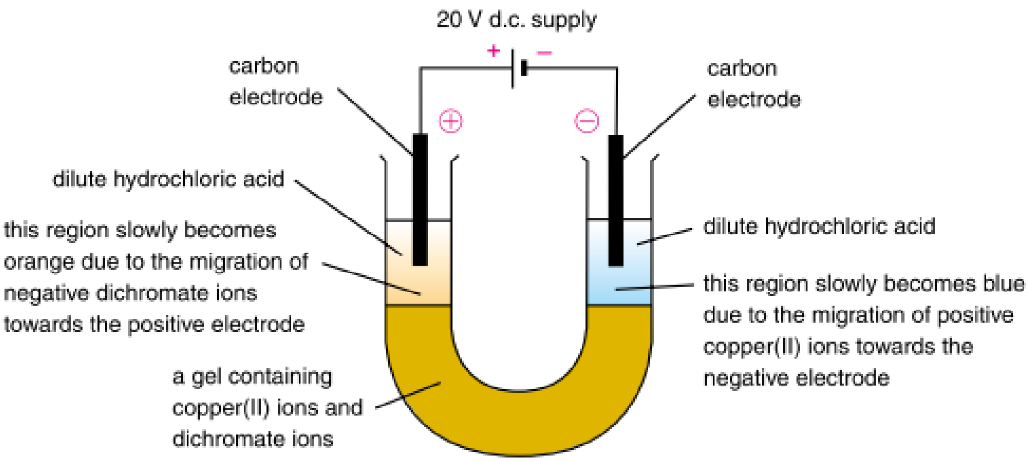
Experimental procedure:

1. Moisten the filter paper with dilute sodium sulphate solution and place it on top of a microscopic slide.
   (Sodium sulphate solution can increase the electrical conductivity of the filter paper by providing mobile ions)
2. Put a small crystal of potassium permanganate crystal at the center on the filter paper
3. Connect each side of the filter paper using electrical wires to the positive and negative pole of the d.c. power supply respectively by crocodile clip.
4. Observe the migration of colored ions after a few minutes. Purple color should move towards positive pole because MnO_4^-, a purple anion, will be attracted by positive pole

Note 1: If the paper is not connected to d.c. supply, the purple color patch will diffuse to all directions.

Note 2: To speed up the experiment, higher voltage can be applied.

Experiment 2: Using U-tube



- Dilute hydrochloric acid can increase the electrical conductivity of the filter paper by providing mobile ions
- Larger amount of compound can be used in this method ➔ Enable easier observation
- The gel can slow down the mixing of the bottom layer in the U-tube with the top layer. It also slows down the migration of Cu^{2+}(aq) and $Cr_2O_7^{2-}$(aq) ions to make the results more easily observed.

Intensive notes (Topic 2: Microscopic world I)

Covalent bond

Covalent bond is the strong directional electrostatic attraction between the shared electrons and the two nuclei of the bonded atoms. It is formed by sharing of outermost shell electrons between two atoms
Molecule is the smallest part of an element or a compound which can exist on its own under room conditions

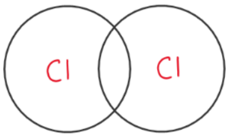
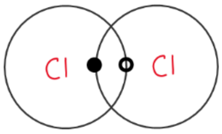
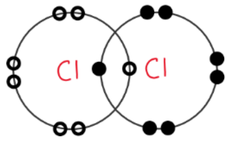
Steps for drawing electron diagrams of covalent molecules

Group	IV	V	VI	VII	0	Hydrogen
No. of electrons needed to attain an octet structure	4	3	2	1	------	1 (duplet)

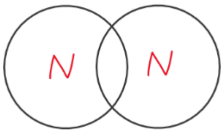
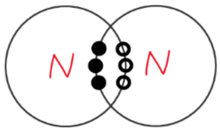
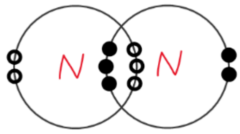
General rule 1: If an atom needs **n** electrons to attain an octet structure, **n** of its outermost shell electrons is shared

General rule 2: In a covalent bond, both atoms contribute equal number of electrons during sharing

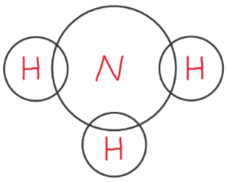
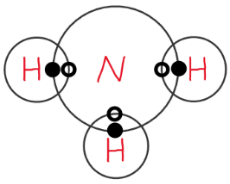
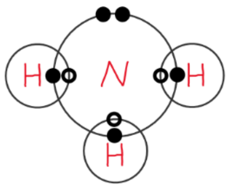
Example 1 Cl_2

Step 1		Step 2		Step 3	
Place 2 Cl atoms		Since Cl needs 1 electron to become octet, each Cl atom shares 1 electron		Fill in the unused electron for each Cl atom (7 – 1 = 6 for each Cl atom)	

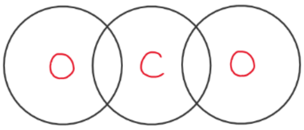
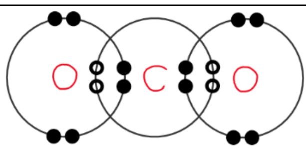
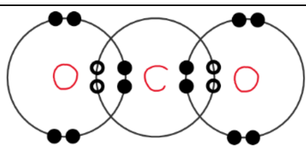
Example 2 N_2

Step 1		Step 2		Step 3	
Place 2 N atoms		Since N needs 3 electron to become octet, each N atom shares 3 electrons		Fill in the unused electron for each N atom (5 – 3 = 2 for each N atom)	

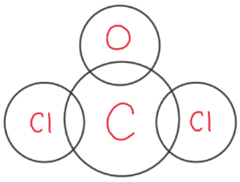
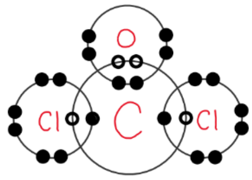
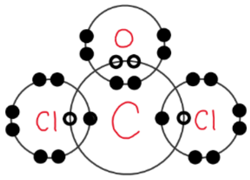
Example 3 NH_3

Step 1		Step 2		Step 3	
Place the N and H atoms (For molecule with AB_X pattern, A is the central atom)		Ensure all non-central atoms become stable (H becomes duplet by sharing 1 electron with N)		Fill in the unused electron for central atom N (5 – 1x3 = 2 for N atom)	

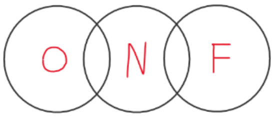
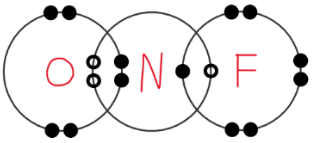
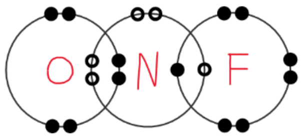
Example 4 CO_2

Step 1		Step 2		Step 3	
Place the C and O atoms (For molecule with AB_X pattern, A is the central atom)		Ensure all non-central atoms become stable (O becomes octet by sharing 2 electrons with C)		Fill in the unused electron for central atom C (All 4 electrons of C are used, no electrons need to be filled)	

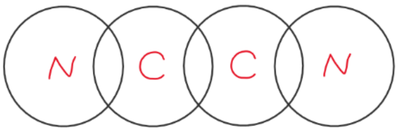
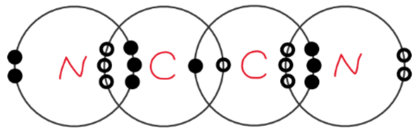
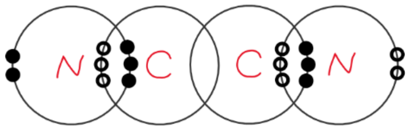
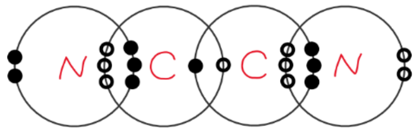
Example 5 **$COCl_2$**

Step 1		Step 2		Step 3	
Place the C, O and Cl atoms (C is the central atom as its group number is lowest)		Ensure all non-central atoms become stable (Cl shares 1 electron with C, O shares 2 electrons with C)		Fill in the unused electron for central atom C (All 4 electrons of C are used, no electrons need to be filled)	

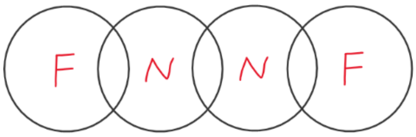
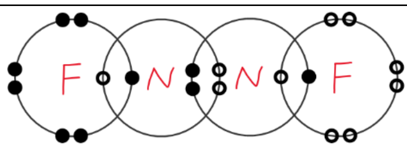
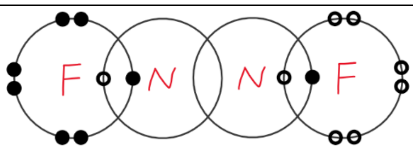
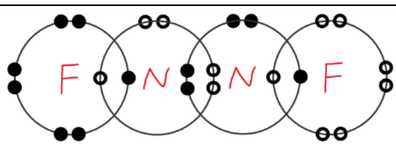
Example 6 **NOF**

Step 1		Step 2		Step 3	
Place the N, O and F atoms (N is the central atom as its group number is lowest)		Ensure all non-central atoms become stable (F shares 1 electron with N, O shares 2 electrons with N)		Fill in the unused electron for central atom N (5 – 2 – 1 = 2 for N atom)	

Example 7 **C_2N_2**

Step 1		Step 3	
Place the C and N atoms (For molecule with A_xB_y pattern, the atoms should be placed symmetrically. Atoms with lowest group number (i.e. C) should be placed in the middle)		Ensure the central atoms forming enough covalent bonds (Each C atom should form 4 bonds. Therefore two C atoms should from 4 – 3 (already formed with N) = 1 bond)	
Step 2		Step 4	
Ensure all non-central atoms become stable (N shares 3 electrons with C to attain an octet structure)		Fill in the unused electron for central atom C (All 4 electrons of C are used, no electrons need to be filled)	

Example 8 **N_2F_2**

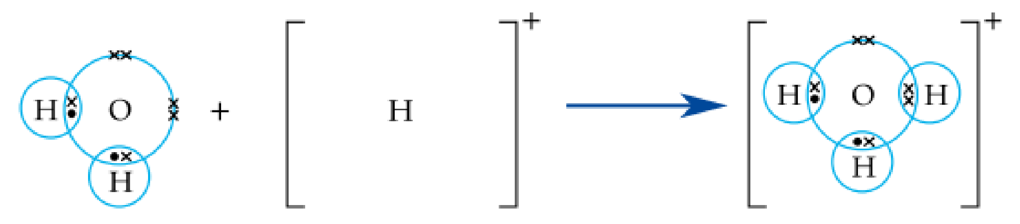
Step 1		Step 3	
Place the N and F atoms (For molecule with A_xB_y pattern, the atoms should be placed symmetrically. Atoms with lowest group number (i.e. N) should be placed in the middle)		Ensure the central atoms forming enough covalent bonds (Each N atom should form 3 bonds. Therefore two N atoms should from 3 – 1 (already formed with F) = 2 bonds)	
Step 2		Step 4	
Ensure all non-central atoms become stable (F shares 1 electron with N to attain an octet structure)		Fill in the unused electron for central atom N (5 – 2 – 1 = 2 for N atom)	

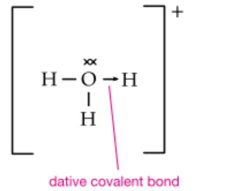
Dative covalent bond

Dative covalent bond is a covalent bond formed between two atoms where both electrons of the shared pair are contributed by the same atom

Example 1: Describe the formation of dative covalent bond in hydronium ion (H_3O^+).

- H_2O has 2 lone pairs of electrons on O atom.
- H^+ does not have any electron in its outermost shell. It can accept 2 more electrons to attain a stable duplet structure.
- Dative covalent bond is formed when H_2O shares its lone pair electrons on O to H^+.

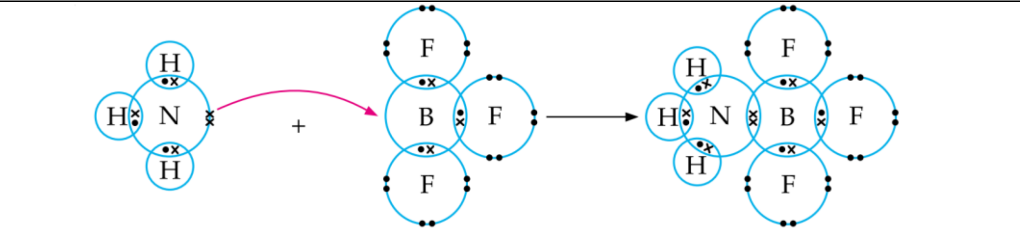


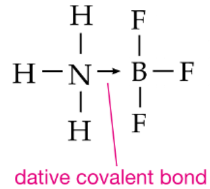


Structural formula of H_3O^+

Example 2: Describe the formation of dative covalent bond in between NH_3 and BF_3.

- NH_3 has a lone pair of electrons on N atom.
- B atom in BF_3 has 6 outermost shell electrons. It can accept 2 more electrons to attain a stable octet structure.
- Dative covalent bond is formed when NH_3 shares its lone pair electrons on N to B atom in BF_3





Structural formula of H_3NBF_3

Other examples:

$NH_3 + H^+ \rightarrow NH_4^+$	$F^- + BF_3 \rightarrow BF_4^-$	CO	$AlCl_3 + AlCl_3 \rightarrow Al_2Cl_6$
$O_2 + O \rightarrow O_3$	$N_2 + O \rightarrow N_2O$	$NO + O \rightarrow NO_2$	$n\ BeCl_2 \rightarrow (BeCl_2)_n$

Name of covalent compounds

1. Element occurs first in the following series. (Formula of water is H_2O, NOT OH_2)

   B, Si, C, P, N, H, S, I, Br, Cl, O, F

2. **Mono (1), di (2), tri (3), tetra (4), penta (5), hexa (6)** can indicate the number of atoms of that element in the compound
3. Name of the second element ends up with **-ide** (e.g. carbon dioxide, not carbon dioxygen)
4. Some hydrogen-containing compounds have special names

   CH_4: Methane **NH_3: Ammonia** **H_2O: Water** **H_2O_2: Hydrogen peroxide**

Summary of chemical bonding

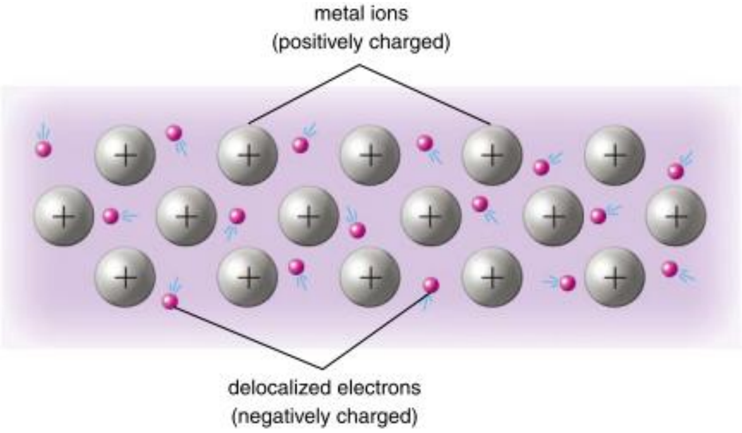
	Ionic bond	Metallic bond	Covalent bond
Nature	**Non-directional** electrostatic attraction between **cations** and **anions**	**Non-directional** electrostatic attraction between **metal ions** and **sea of delocalized electrons**	**Directional** electrostatic attraction between **shared electrons** and **two nuclei of the bonded atoms**
Formation	Transfer of electron(s) from one atom (group of atoms) to another	Escape of outermost shell electrons from metal atoms	Sharing of outermost shell electrons between two atoms

Giant metallic structure

All metals have giant metallic structure. Outermost shell electrons of metal atoms are delocalized

Giant lattice of positive metal ions are surrounded by a sea of delocalized electrons

- Metallic bond is the strong, non-directional electrostatic attraction between the delocalized electrons and metal ions

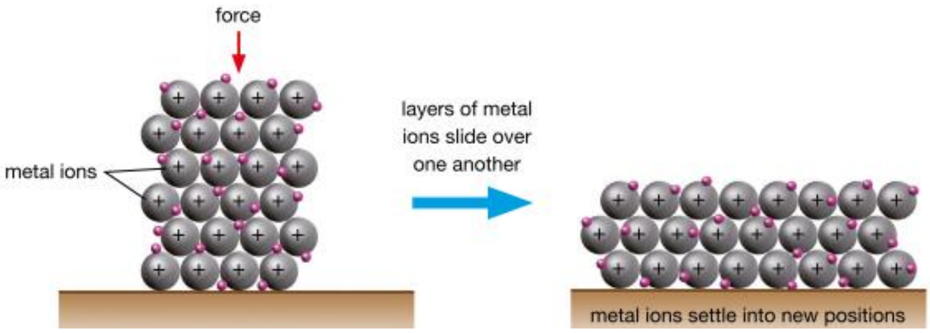


Properties of substances with giant metallic structure:

- Good conductor of heat and electricity ➔ Due to the presence of delocalized electrons
- High melting point ➔ Metal ions and delocalized electrons are held by strong metallic bond
- High density ➔ Metal ions are packed closely

Q: Why are metals malleable and ductile?

A: In a giant metallic structure, the metal ions pack in layers and are held together by metallic bonds. When we apply a force on a piece of metal, the layers of metal ions can slide over one another. Since the non-directional metallic bonds still hold the metal ions together, the ions settle into new positions and the piece of metal takes on a new shape



Q: How do metals conduct electricity?

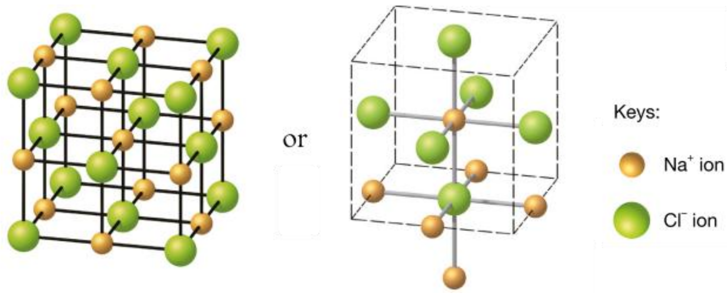
A: In a piece of metal, delocalized electrons move in all directions. When both ends of the metal are connected to a battery, the delocalized electrons move in one direction only.

Q: Why is there no chemical change when metal metals conduct electricity?

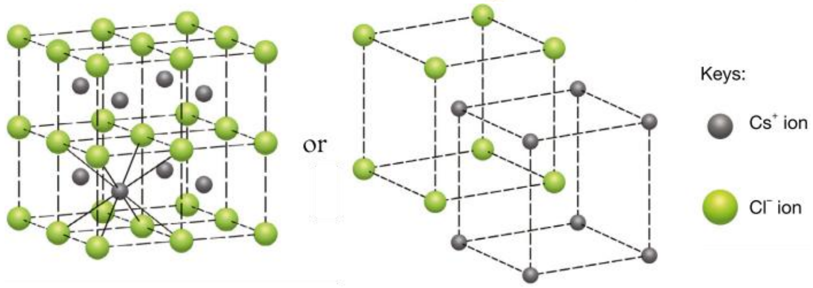
A: Delocalized electrons move towards the positive pole of the battery, leaving the metal. At the same time, equal number of electrons move into the other end of the metal from the negative pole. Number of electrons in the metal remains unchanged at any moment.

Giant ionic structure

Sodium chloride



Caesium chloride

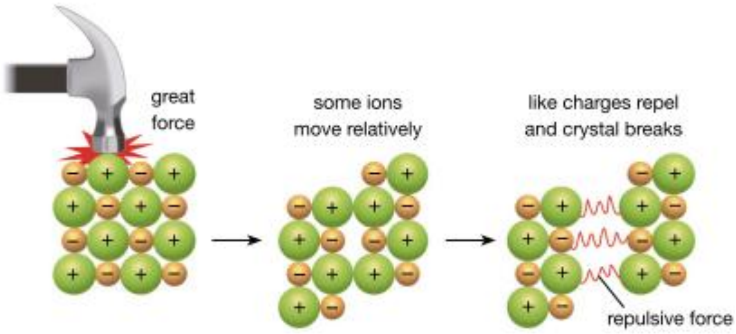


Properties of substances with giant ionic structure:

- Conductor of electricity in aqueous state and molten state ➔ Due to the presence of mobile ions
  (Ions in solid state are NOT mobile. Therefore ionic compounds do not conduct electricity in solid state)
- High melting point ➔ Cations and anions are held by strong ionic bond

Q: Why ionic compounds are brittle?

A: The oppositely charged ions are held together by strong ionic bonds which make the ionic compounds hard. However, when under a great force, the relative movement of the ions brings ions of the same charge close to each other. This will result in repulsion and the crystal will break. Therefore, ionic compounds are brittle.

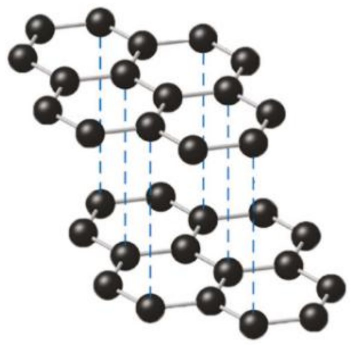


Q: Using sodium chloride as an example, explain why ionic compounds are soluble in water but insoluble in organic solvent

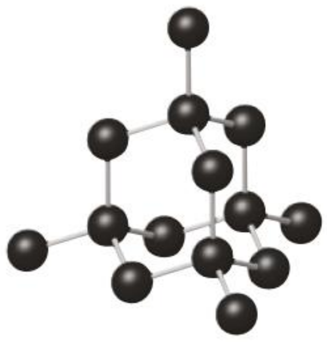
A: When solid sodium chloride is added to water, attraction exists between ions in sodium chloride and water molecules. This attraction causes the sodium ions and chloride ions to move away from the crystal and go into the water. Then water molecules surround the ions. The ions are said to be hydrated. No such attraction exists between the ions in sodium chloride and molecules of organic solvents. Thus, sodium chloride is insoluble in organic solvents.

Intensive notes (Topic 2: Microscopic world I)

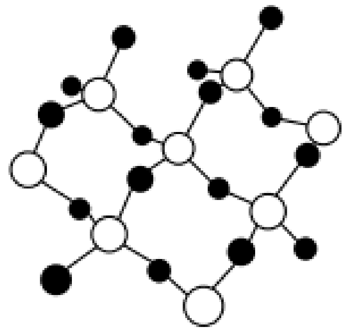
Giant covalent structure



Graphite



Diamond



Silicon dioxide (Quartz)

Properties of substances with giant covalent structure:

- Non-conductor ➔ Absence of delocalized electrons and mobile ions
  (Except graphite which can conduct electricity by delocalized electrons)
- High melting point ➔ Atoms are held by strong covalent bond

Q: Explain why graphite is brittle while diamond is hard

A: In graphite, graphite layers are held together by weak van der Waals' forces. Graphite layers can be separated easily.
In diamond, carbon atoms are held together by strong covalent bonds. The relative movement of atoms is restricted.

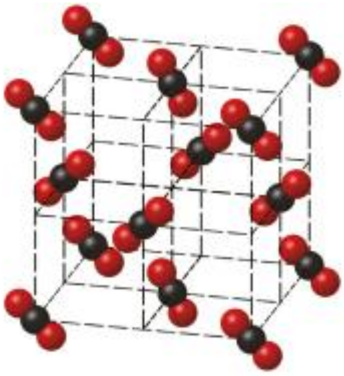
Q: Explain why graphite can conduct electricity while diamond cannot.

A: In graphite, each carbon atom is covalently bonded to 3 other carbon atoms in its layer and 1 outermost shell electron is delocalized. These delocalized electrons can move freely within a graphite layer
In diamond, each carbon atom is covalently bonded to 4 other carbon atoms. There are no delocalized electrons in diamond.

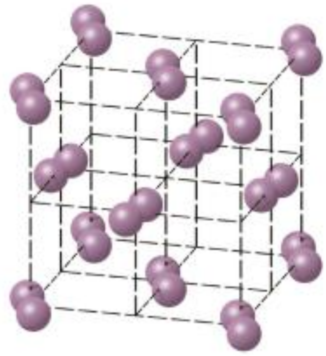
Q: Explain why substances with giant covalent structure cannot dissolve in both polar and non-polar solvents.

A: It is difficult to break the covalent bonds and separate the atoms.

Simple molecular structure



Carbon dioxide



Iodine

Properties of substances with simple molecular structure:

- Non-conductor ➔ Absence of delocalized electrons and mobile ions

  (Except acids (HCl, HNO_3, H_2SO_4) and alkalis (NH_3) which can conduct electricity in aqueous state by mobile ions)

- Low melting point ➔ Molecules are held by weak van der Waals' forces

  Note: When melting substances with simple molecular structures (e.g. methane), only weak van der Waals' forces between molecules are going to overcome. The strong covalent bonds between atoms will NOT be broken.

Q: Explain, in terms of intermolecular forces, why the boiling point of bromine is higher than that of chlorine.

A: Molecular size of bromine is larger than that of chlorine

van der Waals' forces between bromine molecules are stronger than those between chlorine molecules.

Q: Explain why substances with simple molecular structure are brittle

A: As the molecules are held together by weak van der Waals' forces.

Q: Using iodine as an example, explain why substance with simple molecular structure are generally soluble in organic solvents but not water.

A: The attraction between water molecules is quite strong (due to the presence of strong hydrogen bonds). When iodine is added to water, the weak attraction (van der Waals' forces) between iodine and water molecules is NOT strong enough to overcome the attraction between the water molecules. Iodine molecules and water molecules therefore cannot mix together.

In heptane (an organic solvent), heptane molecules are held together by weak attraction (van der Waals' forces).

When iodine is added to heptane (an organic solvent), iodine and heptane molecules can mix together readily.

Intensive notes (Topic 2: Microscopic world I)

Structure and properties of different substances

Structure	Giant metallic	Giant covalent	Giant ionic	Simple molecular
Examples	- All metals	- Graphite - Diamond - Silicon dioxide (SiO_2) - Silicon - Boron	- Metal + non-metal compounds - Ammonium (NH_4^+) compounds	- Non-metals - Non-metal + non-metal compounds - Non-metal + semi-metal compounds
Particles	Atoms (containing metal ions and delocalized electrons)	Atoms	Ions	Molecules (containing various number of atoms)
Bonding	Metallic bond between metal ions and sea of delocalized electrons	Covalent bond between atoms	- Ionic bond between ions - Covalent bond between atoms in polyatomic ions	- Covalent bond between atoms (except noble gases) - van der Waals' forces between molecules
m.p. / b.p.	High (**Metal ions** and **sea of delocalized electrons** are held by **strong metallic bonds**)	High (**Atoms** are held by **strong covalent bonds**)	High (**Catons** and **anions** are held by **strong ionic bonds**)	Low (**Molecules** are held by **weak van der Waals' forces**)
Electrical conductivity	High (in solid / liquid state) due to the presence of **delocalized electrons**	Low (Except graphite due to the presence of **delocalized electrons**)	High (in liquid / aqueous state) due to the presence of **mobile ions**	Low (Except aqueous acids alkalis due to the presence of **mobile ions**)
Solubility	Insoluble in any solvents (except those reacts with water)	Insoluble in any solvent	Usually soluble in polar solvents (water)	Usually soluble in non-polar solvents (organic solvents)

Substances that can conduct electricity:

1. Metals (A conductor, conduct electricity by **delocalized electrons <u>without</u>** decomposition)
2. Graphite (A conductor, conduct electricity by **delocalized electrons <u>without</u>** decomposition)
3. Aqueous / molten salts (An electrolyte, conduct electricity by **mobile ions <u>with</u>** decomposition)
4. Aqueous acids / alkalis (An electrolyte, conduct electricity by **mobile ions <u>with</u>** decomposition)